BLADE-NCO Theory Research
Stages 1–2: Formal Alignment and Mechanistic Reachability

Research note

October 21, 2026

Abstract

This note formalizes structural reachability, inherited population memory, moving hard-instance regions, and first discovery measured in computational cost. It records three candidate claims, attempts a proof of a finite-window memory comparison, and tests broad memory/diversity claims against exact finite-state examples. The strongest surviving result is conditional: retained structural memory helps only when it supplies a verified per-cost hit-hazard advantage over the comparator. No unconditional BLADE-NCO advantage is established. This is a Stage 1 note, not a completed theory paper.

Stage 1.5 corrects the state space, time indices, probability space, aggregation language, and implementation mapping without adding a theorem or changing that conclusion. The finite examples remain counterexamples to universal claims; they are not assumptions that the TSP instance space is finite.

Stage 2 proves a deterministic occupied-niche persistence invariant for the verified parent–child union and Elite-QD selector, while showing by exact counterexample that niche persistence need not preserve useful structural representatives. It also gives a finite-budget multistep bridge-chain bound whose factors are operator-specific accepted-transition and selection-retention certificates. The theorem is mathematically valid under its stated premises; the decisive BLADE-specific kernel and bridge-retention premises remain open.

1 Scope and status

The scope is restricted to structural reachability, evolutionary memory, and expected first-hitting time under a non-stationary hardness landscape. The supplied BLADE-NCO context is FOGA-11 structural mutation, inherited parent populations, HGB40 hardness estimation, Elite-QD1000 (P=T), and BQ solver co-adaptation. Those names are treated as labels, not as complete algorithm specifications.

Statuses are reported on three separate axes. [**Mathematical status: PROVED**], for example, concerns the validity of a mathematical statement under its premises; [**Implementation applicability: VERIFIED, PARTIAL, OPEN, or NOT APPLICABLE**] concerns whether those premises and variables match the inspected S6-AE implementation; and [**Scientific novelty: OPEN or NOT CLAIMED**] concerns literature priority. A proof of a conditional probability bound does not establish implementation applicability or novelty. Compilation verifies only that the source typesets; it does not verify mathematical correctness.

The Stage 1.5–2 implementation boundary is the verified HGB40 + Elite-QD1000 ($P = T$) S6-AE path. Its frozen artifacts passed 40/40 strict checks. Historical GP4 output is not used as the definition of the current method. Details that could not be established from the inspected code and artifacts are marked OPEN rather than inferred.

2 Prior-theory boundary

Related theory includes rigorous results in which populations track moving optima and in which diversity mechanisms have different effects, but those works analyze explicit dynamic-function classes and specified algorithm dynamics [1, 2]. MAP-Elites and QD explicitly store high-quality representatives in a fixed feature partition [3, 4]; recent runtime analyses prove coverage and optimization results for particular pseudo-Boolean and combinatorial structures [5, 6]. These works make the elementary archive-occupancy invariant non-novel, and they show that stepping-stone benefits require an aligned problem/operator model. They do not establish compute-normalized discovery of solver-relative hard TSP instances under surrogate selection and solver co-adaptation. Whether a fully instantiated BLADE-specific bridge theorem would be novel remains **[Scientific novelty: OPEN]**; the present limited audit is not a comprehensive priority search.

3 Formal model and implementation boundary

3.1 Structural states and mutation

Let $\mathcal{Z}$ be the set of encoded candidate instances and let $\phi : \mathcal{Z} \to \mathcal{X}$ be a measurable map from encodings to the solver-visible TSP instance. For the inspected fixed-size experiments one may take $\mathcal{X}$ to be the valid canonical subset of $[0,1]^{100\times 2}$, equipped with its Borel σ-algebra $\mathcal{B}_{\mathcal{X}}$; a variable-size model may instead use a measurable disjoint union over instance sizes. Encodings z, z' are equivalent when $\phi(z) = \phi(z')$. A structural state is a phenotype $x \in \mathcal{X}$, for example a canonical graph/instance together with whichever structural attributes determine solver input. The exact canonicalization and attributes are implementation questions. We assume ϕ and every event used below are measurable.

The instance space is not made finite by a finite search budget. A realized finite-budget run contains finitely many attempted instances, but the support of a continuous-coordinate mutation kernel can remain uncountable. For each BQ cycle k and FOGA operator o, adjoin an isolated rejection symbol ∂ and write $\overline{\mathcal{X}} = \mathcal{X} \dot\cup \{\partial\}$ with the natural extended sigma-algebra. Let

$$\widetilde{K}_{k,o}(x, \cdot \mid s)$$

be a Markov kernel from a parent x and full pre-mutation state s into $\overline{\mathcal{X}}$. An outcome in $\mathcal{X}$ is the valid repaired phenotype of an accepted proposal; ∂ records a rejected or invalid proposal. The kernel includes stochastic operator parameters, repair, and canonicalization. Dependence on k or s permits solver-, archive-, or history-dependent behavior. The corresponding encoding process need not remain Markov after projection through ϕ.

Let $d : \mathcal{X} \to \mathbb{R}^4$ be the measurable QD descriptor map, including the frozen normalization, and let $g : \mathcal{X} \to \{1, \ldots, K\}$ be nearest-centroid assignment with a declared measurable tie rule. Then

$\mathcal{N}_j = g^{-1}(\{j\})$ gives a finite measurable partition after discarding any empty cells. Finiteness belongs to the niche index set, not to $\mathcal{X}$.

3.2 Two time scales and the S6-AE batch

Use k only for the BQ adaptation cycle. At the start of cycle k, the BQ solver state q_k and the parent population $P_k = (X_{k,1}, \ldots, X_{k,m})$ are fixed, with $m = 10{,}000$ in S6-AE. Let P_k also denote its membership set when used inside $\subseteq$, $\cap$, or $\mathsf{S}(\cdot)$; the ordered tuple is used for scheduling. Let $\mathcal{O}$ be the ordered list of the 11 FOGA operators. The within-cycle mutation attempts are indexed by $(i, o) \in \{1, \ldots, m\} \times \mathcal{O}$:

$$Z_{k,i,o} \sim \widetilde{K}_{k,o}(X_{k,i}, \cdot \mid S_k).$$

This notation denotes a conditional law, not random parent or operator selection. The inspected implementation uses the deterministic Cartesian schedule "parents in stored order, then operators in registry order," so each parent–operator pair is attempted exactly once. A content-derived seed from $(k, \mathrm{hash}(X_{k,i}), o)$ drives the stochastic operator parameters and repair. If $Z_{k,i,o} = \partial$, that scheduled attempt produces no child and is not repeated. Conditioning the kernel on acceptance would instead describe an implicit-resampling algorithm and is not used here.

Let FU retain first occurrences in a finite ordered list by instance hash. Apply it to the subsequence of $(Z_{k,i,o})_{i,o}$ lying in $\mathcal{X}$ to obtain the accepted, child-deduplicated list O_k. An accepted duplicate is removed by this deterministic map; it is not redrawn. Let $C_k = \mathrm{unique}(P_k \cup O_k)$ be the candidate union, retaining the first occurrence under the declared order. At C00 S6-AE reuses the hash-locked corrected union; later cycles construct this union directly. The cycle then fits/scores with HGB40, selects the acquisition/next-parent set, obtains exact labels for that selected set, and only then adapts the BQ solver from q_k to q_{k+1}. Thus within-cycle mutation attempts and solver cycles must not share one index.

3.3 Population memory and parent survival

At BQ cycle k, let P_k be the ordered parent population and A_k the branch-local labeled archive. Let S_k collect scores, lineage information, solver state, and other algorithm variables that affect the next transition. Thus $M_k = (P_k, A_k, S_k)$ is the operational memory state. Parent survival is a transition event: for parent i, its survival probability is

$$r_{k,i} = \mathbb{P}(X_{k,i} \in P_{k+1} \mid \mathcal{F}_k^{\mathrm{pre}}).$$

In a generic policy this may differ from a proposal-selection probability. For S6-AE, however, no such random parent-selection variable is present: every member of P_k receives all 11 scheduled mutations. Population update is a deterministic function of the realized candidate union, HGB40 scores, niche assignments, and hash tie-breaks. Randomness can still enter through the roots, mutation parameters, calibration sample, model fitting, training permutations, or nondeterministic numerical execution. The full process may be time-inhomogeneous Markovian in a sufficiently rich (M_k, θ_k, q_k); a niche label or count vector alone need not be.

For a finite candidate set A, define its occupied-niche set by

$$\mathsf{S}(A) = \{g(x) : x \in A\}.$$

For a finite unique candidate union C and score $\widehat{h}$, let $e_j(C)$ be the maximum-score member of $C \cap \mathcal{N}_j$, using ascending instance hash to break ties, and define the per-niche floor

$$Q(C) = \{e_j(C) : j \in \mathsf{S}(C)\}.$$

This set has $|\mathsf{S}(C)|$ members, not 10,000 members in general. If the population capacity is m, $|\mathsf{S}(C)| \le m \le |C|$, let $F_m(C)$ be the first $m - |Q(C)|$ candidates of $C \setminus Q(C)$ under global descending $\widehat{h}$ and ascending-hash tie order. The complete selected population is

$$T_m(C) = Q(C) \cup F_m(C).$$

The notation denotes the membership set; the implementation also records a deterministic order.

In verified S6-AE, $m = 10{,}000$, $K = 1000$, $Q_k = Q(C_k)$, and $T_k = T_m(C_k)$. The result has 10,000 unique members and is used both as the BQ acquisition set and next parent population: $P_{k+1} = T_k$. "Elite-QD1000" therefore names the occupied-niche floor Q_k, not the full T_k. The score can change at every cycle without changing the fixed partition g.

For any $x \in C$, the deterministic membership criterion is

$$x \in T_m(C) \quad \Longleftrightarrow \quad x = e_{g(x)}(C) \text{ or } \left[x \notin Q(C) \text{ and } \operatorname{rank}_{C \setminus Q(C)}(x) \le m - |Q(C)|\right].$$

Consequently, retaining a niche does not imply retaining any particular parent hash in that niche.

3.4 Time-indexed bridge sets

For a prospective path starting at cycle k, let $B_0, \ldots, B_L \in \mathcal{B}_{\mathcal{X}}$ be measurable bridge sets, where B_ℓ is intended to be represented in $P_{k+\ell}$ and $B_L \subseteq \mathcal{H}_{k+L-1}$ is a final true-hard set generated during cycle $k + L - 1$. Define

$$E_\ell = \{P_{k+\ell} \cap B_\ell \neq \varnothing\}, \qquad G_\ell = \{C_{k+\ell} \cap B_{\ell+1} \neq \varnothing\}.$$

The sets may occupy different niches and may depend on the declared future solver-state sequence. They are structural certificates, not HGB threshold sets.

For a population context and horizon, a future-reachability value may be written

$$V_{k,L}(x) = \mathbb{P}(\text{a declared future true-hard set is hit within } L \text{ cycles} \mid x \text{ is retained at cycle } k,\ \mathcal{F}_k).$$

This quantity generally depends on the other population members, future selection, and solver states. It is not a function of current true hardness $h_k^{\mathrm{BQ}}(x)$ or current score $\widehat{h}_k(x)$ unless an additional alignment assumption is proved.

3.5 Probability space and frozen executions

Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry exactly the random inputs declared for the analysis: possibly random initial roots, content-derived operator seeds and their stochastic parameters/repair, seeded archive calibration sampling, HGB estimator seeds, BQ training permutations and suffix sampling, and any explicit selection randomness in a comparator. Conditioning on the frozen S6-AE roots, source artifacts, software/runtime, and all recorded seeds gives a single realized trajectory ω_0; many nominal "probabilities" then collapse to zero or one. A probabilistic reachability theorem must therefore state which inputs vary across Ω. Repeatedly serializing a deterministic batch does not create random parent/operator selection, and unspecified hardware nondeterminism is not silently treated as a scientific sampling law.

3.6 Hardness and moving target

Let θ_k describe the task/environment and q_k the state of the co-adapting BQ solver before cycle-k adaptation. Define true BQ hardness $h_k^{\mathrm{BQ}}(x) = h(x; \theta_k, q_k)$, where a larger value means harder under a prespecified solver-performance metric. For threshold κ, the true hard region at cycle k is

$$\mathcal{H}_k = \{x \in \mathcal{X} : h_k^{\mathrm{BQ}}(x) \geq \kappa\}.$$

This definition allows the target to move because either the task or solver state changes. During the S6-AE mutation/acquisition part of cycle k, q_k is fixed; adaptation produces q_{k+1} only after T_k is frozen and labeled. For the proofs, the target attached to a declared opportunity must be predictable from its pre-opportunity history.

HGB40 supplies a cycle-specific prediction $\widehat{h}_k(x)$ (in the implementation, predicted $\log(1 + \mathrm{LBQ})$), not the definition of h_k^{BQ} or $\mathcal{H}_k$. Define the latent generation-level event

$$D_k^{\mathrm{gen}} = \{\exists x \in C_k : h_k^{\mathrm{BQ}}(x) \geq \kappa\},$$

and the acquisition-level event

$$D_k^{\mathrm{acq}} = \{\exists x \in T_k : h_k^{\mathrm{BQ}}(x) \geq \kappa\}.$$

The former concerns what the generation-level candidate union contains, including retained parents and accepted children, even when exact hardness was not measured; the latter concerns what survived HGB/QD acquisition and can be verified. They are distinct from the event $\{\widehat{h}_k(x) \geq \widehat{\kappa}\}$ and from post-adaptation hardness $h_{k+1}^{\mathrm{BQ}}(x)$. Without a calibration/ranking-and-retention condition, none of these quantities can be substituted for another.

3.7 First hit and computational normalization

For the abstract survival lemma, index declared comparison opportunities by j; j is not automatically a BQ cycle. An opportunity may be one serialized mutation attempt, one full generation event D_k^{gen}, or one acquisition event D_k^{acq}, but the choice must be fixed before the comparison. Let

$$\tau = \inf\{j \geq 1 : Y_j = 1\}, \qquad C_\tau = \sum_{j=1}^{\tau} c_j,$$

where Y_j is the declared true-hardness discovery event and $c_j > 0$ is the normalized compute cost through resolving opportunity j. The budget-B discovery event is $\{C_\tau \leq B\}$. The cost must include the work that differs between methods (structural generation, HGB40 inference, archive operations, and BQ evaluation/verification, as applicable). Counting generations alone is not computational normalization. The elementary bound below uses deterministic opportunity costs c_i; random or method-dependent costs require a cost-aware hazard analysis and are not silently replaced by generation counts.

For a policy π, define its conditional hit hazard

$$q_j^\pi = \mathbb{P}(Y_j = 1 \mid \mathcal{F}_{j-1}^\pi),$$

where $\mathcal{F}_{j-1}^\pi$ is the complete pre-opportunity history. On a no-hit history, a lower bound on q_j^π is a reachability certificate. A Markov state representation is one way to calculate this hazard, but the history-based definition does not require Markovianity.

3.8 Abstraction versus verified BLADE-NCO context

Supplied component	Representation here	Unspecified for a BLADE-specific proof
FOGA-11 mutation	Eleven extended kernels $\widetilde{K}_{k,o}$ on $\overline{\mathcal{X}}$	Exact kernel laws over stochastic parameters, rejection, and support bounds
Inherited parents; Elite-QD1000 (P=T)	P_k, measurable partition g, deterministic HGB/QD selection, $P_{k+1} = T_k$	Distributional survival/reachability bounds under a declared ensemble
HGB40 and BQ co-adaptation	Prediction $\widehat{h}_k$, true BQ hardness h_k^{BQ}, solver state q_k	Scientific threshold, calibration/retention bridge, and matched validation cost
Fixed generator and HAC	Comparator proposal kernels	Exact proposal laws, gradient access, stopping rules, and per-opportunity compute
Finite budget	Cumulative cost C_τ	Which compute resources are charged and how variable costs are normalized

Accordingly, the formal model is a deliberately explicit abstraction. The verified schedule and selection mapping do not assert unobserved transition probabilities. The remaining implementation-to-theory gaps are recorded as open assumptions in `assumption_register.md`.

3.9 Niche aggregation and the Markov condition

Suppose a Markov kernel Q on a full state space is aggregated by a finite measurable partition $\{B_1, \ldots, B_L\}$. The strong-lumpability equalities

$$Q(s, B_j) = Q(s', B_j) \quad \text{for every } s, s' \in B_i \text{ and every } i, j$$

are sufficient for the aggregate block process to be Markov for every initial law. They are not necessary for a particular initial law. A weaker, distribution-dependent Markov aggregation may hold when the conditional within-block law induced by that initial distribution and history makes the averaged transition to each B_j history-independent (possibly with a time-dependent aggregate kernel). Such a claim must name that law and prove the conditional equality; it is not obtained merely by averaging once.

The counterexample in Section 6.1 shows failure of the strong condition and therefore invalidates an implementation-independent Markov claim based on niche alone. It does not rule out a special weak aggregation for a specified initial distribution. In S6-AE, niche summaries hide parent identity, within-niche HGB order, candidate age, solver/archive state, and the deterministic hash tie-break, so strong lumpability is not established and a distribution-dependent alternative remains OPEN.

4 Candidate propositions and Stage 2 results

4.1 Stage 2 Proposition: deterministic niche persistence

[Mathematical status: PROVED] [Implementation applicability: VERIFIED for S6-AE] [Scientific novelty: NOT CLAIMED]

Proposition 1 (Occupied-niche persistence under the verified selector)**.** *Fix a niche map g. At cycle k, suppose (i) every member of P_k is in C_k; (ii) C_k is finite and unique by instance hash; (iii) $|\mathsf{S}(C_k)| \le m \le |C_k|$; (iv) $T_k = T_m(C_k)$ contains the complete per-niche floor $Q(C_k)$; and (v) $P_{k+1} = T_k$. Then*

$$\mathsf{S}(P_{k+1}) = \mathsf{S}(C_k) \supseteq \mathsf{S}(P_k).$$

If $C_k = \mathrm{unique}(P_k \cup O_k)$ under the same fixed map, then more precisely

$$\mathsf{S}(P_{k+1}) = \mathsf{S}(P_k) \cup \mathsf{S}(O_k).$$

The statement remains true when some or all scheduled mutation attempts are rejected, because rejection contributes no child and causes no resampling.

The assumptions match the inspected S6-AE selection path: the corrected C00 union and later parent–child unions contain all old parents; QD4, normalization, centroids, and tie rules are fixed; at most 1,000 niches are occupied while $m = 10{,}000$; selection contains exactly one current-HGB elite from every occupied candidate niche; and $P = T$. The completed run's strict checks verify these operational invariants. This proves a niche-set invariant, not parent-hash survival, within-niche structural equivalence, or future reachability.

Lemma 1 (Exact membership test for a candidate)**.** *Under the deterministic selector in the formal model, a candidate survives if and only if it is its niche's score elite (including the declared hash tie rule), or it lies within the first $m - |Q(C)|$ positions of the global order on $C \setminus Q(C)$. Hence neither membership in P_k nor occupancy of its niche is sufficient for that candidate to survive.*

4.2 Stage 2 theorem candidate: a certified bridge chain

[Mathematical status: PROVED under explicit premises] [Implementation applicability: PARTIAL/OPEN] [Scientific novelty: NOT CLAIMED]

Proposition 2 (Finite-budget multistep bridge-chain bound)**.** *Fix a starting cycle k, bridge sets $B_0, \dots, B_L$, and the complete history filtration at the pre-generation and post-generation/pre-selection points of each cycle. Suppose E_0 holds surely. For $\ell = 0, \dots, L-1$, on every admissible history on which the bridge chain has survived through E_ℓ, assume*

$$\mathbb{P}(G_\ell \mid \mathcal{F}^{\mathrm{pre}}_{k+\ell}) \ge a_\ell.$$

For $\ell = 0, \dots, L-1$, after a generated bridge is present, assume

$$\mathbb{P}(E_{\ell+1} \mid \mathcal{F}^{\mathrm{gen}}_{k+\ell}, G_\ell) \ge \rho_\ell,$$

where $E_{\ell+1}$ means that selection retains a member of $B_{\ell+1}$; for the final cycle this condition is needed only for acquisition-level discovery. Then

$$\mathbb{P}(D^{\mathrm{gen}}_{k+L-1} \text{ by the certified chain}) \geq \prod_{\ell=0}^{L-1} a_\ell \prod_{\ell=0}^{L-2} \rho_\ell,$$

$$\mathbb{P}(D^{\mathrm{acq}}_{k+L-1} \text{ by the certified chain}) \geq \prod_{\ell=0}^{L-1} a_\ell \rho_\ell.$$

No independence is required. If the deterministic cycle costs $c_k, \ldots, c_{k+L-1}$ sum to at most B, these are finite-budget discovery bounds.

A sufficient operator-specific source for a_ℓ is a designated $o_\ell \in \mathcal{O}$ such that every admissible retained $x \in B_\ell$ satisfies

$$\widetilde{K}_{k+\ell,o_\ell}(x, B_{\ell+1} \mid S_{k+\ell}) \geq a_\ell.$$

Because all 11 operators are scheduled once for every parent, no random parent/operator-selection factor appears. Rejection mass at ∂ is already included in the kernel and is not conditioned away.

The selection premise is identifiable rather than an unexplained hit-hazard advantage. For example, $\rho_\ell = 1$ follows if, for every admissible candidate union containing $B_{\ell+1}$, either the deterministic score elite of a designated occupied niche belongs to $B_{\ell+1}$, or a bridge member passes the exact global-fill rank test in Lemma 1. A stronger but easier sufficient condition is within-niche transition equivalence: every possible score elite in that niche has the same certified next-bridge lower bound. None of these score–reachability alignment conditions follows from current-hardness accuracy alone.

For S6-AE, the all-operator schedule, rejection handling, capacity inequality, selector form, and $P = T$ map are verified. Positive, history-uniform a_ℓ values, scientifically meaningful bridge sets, and deterministic or probabilistic ρ_ℓ bounds are OPEN. Changing BQ checkpoints are allowed because the bridge sets and kernels are cycle-indexed, but their alignment must be proved separately at every step.

Repeated certified windows. **[Mathematical status: PROVED under regeneration]** **[Implementation applicability: OPEN]** **[Scientific novelty: NOT CLAIMED]** If after every no-hit L-cycle block the next block again starts with E_0 and satisfies the same conditional premises, set $p = \prod a_\ell \prod \rho_\ell$ for the desired discovery level and let C bound a block's charged compute. The standard survival argument gives $\mathbb{P}(C_\tau \leq rC) \geq 1 - (1-p)^r$ and $\mathbb{E}[C_\tau] \leq C/p$. The regeneration premise is not verified for S6-AE; without it, the one-block finite-budget bound is the strongest conclusion.

4.3 Mechanism-level comparator boundary

[Mathematical status: CONDITIONAL FRAMEWORK] **[Implementation applicability: OPEN]** **[Scientific novelty: OPEN]** The bridge theorem permits a fair structural comparison only after each method receives its own transition and cost model.

- A reset evolution strategy can erase intermediate bridge sets between rounds, so a long path certificate may fail, but a reset distribution can still have larger direct mass on the true-hard region.

- A stationary generator has no inherited structural chain; its discovery law is determined by its sampling mass on the cycle-specific hard region and its per-sample cost. It can beat BLADE when that mass is sufficiently large.
- A HAC-style method with random restarts requires a kernel on its full gradient state, a restart law, stopping rules, and charged forward/backward costs. One gradient step is not equated with one FOGA attempt or one full 110,000-attempt generation. The Stage 1 path counterexample shows only that a useful gradient can win in a toy model; it is not a HAC superiority theorem.

Proposition 1 is a structural-coverage statement. Proposition 2 concerns true-hardness generation or acquisition only when its final bridge set lies in the declared true BQ hard region and, for acquisition, survives selection. Neither QD occupancy nor HGB score alone supplies that implication.

4.4 Candidate 1: finite-window reachability comparison

[Mathematical status: CONDITIONAL] [Implementation applicability: PARTIAL] [Scientific novelty: NOT CLAIMED] The following is the strongest surviving candidate.

Proposition 3 (Memory advantage on a stable target window)**.** *Suppose the true hard region attached to the declared discovery event is constant, $\mathcal{H}_j = \mathcal{H}$, for a window of W comparison opportunities and each opportunity has the same deterministic compute cost $c > 0$. On every no-hit history in that window, suppose a memory policy M has conditional hit hazard at least p_M, and a comparator G has conditional hit hazard at most p_G, with $p_M > p_G$. Then, for $1 \le n \le W$,*

$$\mathbb{P}(\tau_M \le n) \ge 1 - (1 - p_M)^n > 1 - (1 - p_G)^n \ge \mathbb{P}(\tau_G \le n),$$

and, for $2 \le n \le W$,

$$\mathbb{E}[c \min(\tau_M, n)] \le c \sum_{j=0}^{n-1} (1 - p_M)^j < c \sum_{j=0}^{n-1} (1 - p_G)^j \le \mathbb{E}[c \min(\tau_G, n)].$$

For a generic policy that actually samples a parent, one sufficient derivation of p_M is the following: on each no-hit history, with conditional probability at least r a useful retained parent is available; given that event it is selected with probability at least γ; conditional on that selection, its mutation/evaluation reaches the true current hard region with probability at least α. Then $p_M \ge r\gamma\alpha$. No independence among these events is assumed; the bounds are conditional in the stated order.

That random-selection factorization is not the S6-AE schedule. For the verified deterministic Cartesian batch, every retained parent receives all 11 operators. If r lower-bounds availability of a useful parent and α_{batch} lower-bounds, under the explicitly declared RNG ensemble, the conditional probability that its scheduled operator batch generates the declared hit, then the appropriate sufficient bound is $p_M \ge r\alpha_{\text{batch}}$. With roots and all seeds conditioned to their frozen values, this is a statement about a realized batch, not an estimated random parent/operator-selection probability.

Importance. It turns a structural-memory claim into a finite-budget, equal-cost comparison that explicitly accounts for parent access and mutation reachability. It also makes clear that survival alone is not a sufficient mechanism.

Assumptions and realism. A piecewise-stable target window and equal-cost opportunities are useful first approximations. In S6-AE the BQ checkpoint is fixed during mutation/acquisition in cycle k and changes only after T_k is frozen and labeled, but comparison windows spanning adaptations can still move $\mathcal{H}_k$. HGB40, archive, exact-label, and BQ-training costs also differ from a single mutation attempt. The deterministic 10,000-by-11 schedule and $P = T$ update are verified; the history-uniform $r\alpha_{\text{batch}}$ bound, scientific threshold, and matched comparator cost/hazard remain [**Implementation applicability: OPEN**].

Proof approach. Induction on the no-hit survival probability, then the identity expressing expected truncated hitting time as the sum of survival probabilities. It requires no conditional independence and no Markov approximation. Proof details are in `proof_attempts.tex`.

Counterexample and HAC distinction. If memory's conditional hazard is lower because its retained parents are stale, or if a gradient proposal has larger hazard per compute unit, the premise fails. The exact examples in `counterexamples.tex` show both. Thus this is a conditional criterion that can distinguish BLADE from fixed generation or HAC only after their hazards and costs are measured; it does not establish that BLADE wins.

4.5 Candidate 2: inherited memory never hurts

[**Mathematical status: DISPROVED**] [**Implementation applicability: NOT APPLICABLE**] [**Scientific novelty: NOT CLAIMED**] The proposed blanket claim is that retaining historical parents cannot increase first-hitting cost relative to a fixed generator. This is false even with two structural states and one target change. The stale-memory example in Section 6.1 has expected cost 100 for inherited memory versus 2 for a fixed uniform generator. The failure is not a technical gap in Candidate 1: the required hazard dominance is reversed.

Importance and proof approach. A universal no-harm theorem would have supported the research objective directly, but the exact counterexample rules it out. No proof is possible without additional target-drift, forgetting, or reachability assumptions. These extra assumptions would need to be stated and independently justified; they cannot be inferred from the word "memory."

Assumptions and realism. The universal claim would require memory never to reduce true-hardness reachability per unit cost for any target shift or parent-selection history. That is not guaranteed by inherited populations and is contradicted by the stale-memory example; realism for BLADE-NCO is low absent explicit target-aware forgetting and verified hazard bounds.

Strongest proof approach. An exact finite-state counterexample proves the universal claim false. A positive restricted result would need additional drift and reachability premises, which are instead stated explicitly in Candidate 1.

HAC distinction. Nothing in this false claim compares BLADE to HAC. HAC can outperform a poorly aligned retained population, as the path example in Section 6.1 demonstrates.

4.6 Candidate 3: diversity always improves discovery

[**Mathematical status: DISPROVED**] [**Implementation applicability: NOT APPLICABLE**] [**Scientific novelty: NOT CLAIMED**] The proposed blanket claim is that preserving a more diverse population always reduces first-hitting time. In a three-state model, a diverse parent selector spends half of each equal-cost proposal on an irrelevant niche and has expected hitting cost 2, while a focused selector hits in cost 1. The exact construction is in Section 6.1.

Importance and assumptions. Diversity can preserve useful future routes, but its benefit depends on niche relevance, selection allocation, target drift, and cost. A monotonicity proof from diversity cardinality alone cannot work. This counterclaim concerns a universal guarantee, not a claim that Elite-QD1000 behaves like the toy selector.

Realism and strongest proof approach. No selection-allocation or niche-relevance condition is supplied for Elite-QD1000, so the premise is not verified for BLADE-NCO. The exact three-state construction is a complete disproof of the universal claim; a positive diversity theorem would require explicit target-access conditions. This does not rank BLADE against HAC.

HAC distinction. It provides no general ranking against HAC. With equal compute, a focused or gradient-guided policy may spend more proposals on the current reachable hard region; the actual ranking depends on transition probabilities and costs, neither of which is specified here.

5 Proof attempt for the strongest candidate

5.1 History-based survival lemma

[**Mathematical status: PROVED**] [**Implementation applicability: PARTIAL**] [**Scientific novelty: NOT CLAIMED**] The lemma below is a standard finite-horizon hitting bound, included as the proof engine rather than as a novelty claim.

Let Y_j be the indicator of the declared discovery event at comparison opportunity j and let $\tau = \inf\{j \geq 1 : Y_j = 1\}$. The event $\{\tau \geq j\}$ is determined by the history before opportunity j. Assume there are deterministic numbers $p_j \in [0, 1]$ such that, on every pre-opportunity history with no previous hit,

$$\mathbb{P}(Y_j = 1 \mid \mathcal{F}_{j-1}) \geq p_j.$$

This is a history-uniform conditional hazard condition. It allows the target attached to the event to vary with j. The index j is deliberately not the BQ cycle index k: it may represent a serialized mutation attempt, a full generation event D_k^{gen}, or an acquisition event D_k^{acq}, provided the choice and filtration are fixed.

Lemma 2 (Non-stationary survival and cost bound)**.** *For every $n \geq 1$,*

$$\mathbb{P}(\tau > n) \leq \prod_{i=1}^{n} (1 - p_i).$$

If opportunity costs c_i are deterministic, then

$$\mathbb{E}[C_\tau] \le \sum_{i\ge 1} c_i \prod_{j=1}^{i-1}(1-p_j),$$

with the inequality valid also when the right side is infinite. For a budget B, assume $\sum_{i=1}^{\infty} c_i = \infty$ and let $n(B) = \max\{n \ge 0 : \sum_{i=1}^{n} c_i \le B\}$, with empty sum zero.

$$\mathbb{P}(C_\tau \le B) \ge 1 - \prod_{i=1}^{n(B)} (1-p_i).$$

Proof. Write $S_i = \{\tau \ge i\}$ for survival to the start of opportunity i. Since $S_i \in \mathcal{F}_{i-1}$,

$$\begin{aligned}\mathbb{P}(S_{i+1}) &= \mathbb{E}[\mathbf{1}_{S_i}\mathbb{P}(Y_i = 0 \mid \mathcal{F}_{i-1})] \\ &\le (1-p_i)\mathbb{P}(S_i).\end{aligned}$$

Starting from $\mathbb{P}(S_1) = 1$ and iterating gives $\mathbb{P}(S_{n+1}) \le \prod_{i=1}^{n}(1-p_i)$, which is the first claim. For the cost, use the nonnegative-sum identity

$$C_\tau = \sum_{i\ge 1} c_i \mathbf{1}_{\{\tau \ge i\}}, \qquad \mathbb{E}[C_\tau] = \sum_{i\ge 1} c_i \mathbb{P}(S_i),$$

and apply the survival bound termwise. Finally, completion of any of the first $n(B)$ opportunities costs at most B, so budget success is at least the probability of a hit among those opportunities. The first claim supplies the stated lower bound. □

5.2 Deriving the window comparison

For Candidate 1, apply the survival lemma on the W-opportunity stable-target window. The memory policy satisfies $p_j = p_M$ and the comparator satisfies an upper hazard bound $q_j^G \le p_G$ on every no-hit history. The same induction, with the inequality reversed for G, yields

$$\mathbb{P}(\tau_M > j) \le (1-p_M)^j, \qquad \mathbb{P}(\tau_G > j) \ge (1-p_G)^j.$$

Therefore, for $n \le W$,

$$\mathbb{P}(\tau_M \le n) \ge 1-(1-p_M)^n > 1-(1-p_G)^n \ge \mathbb{P}(\tau_G \le n).$$

For $2 \le n \le W$, using $\mathbb{E}[\min(\tau, n)] = \sum_{j=0}^{n-1} \mathbb{P}(\tau > j)$ and multiplying by the common opportunity cost c gives the strict truncated compute-cost comparison in Proposition 3. At $n = 1$ both policies pay the first opportunity cost in the truncated-cost convention, so only the success-probability comparison is strict there.

5.3 Stage 2 proof of deterministic niche persistence

Proof of Proposition 1. For every $j \in \mathsf{S}(C_k)$, the intersection $C_k \cap \mathcal{N}_j$ is nonempty, so $e_j(C_k)$ exists. By construction, $e_j(C_k) \in Q(C_k) \subseteq T_k = P_{k+1}$. Hence $\mathsf{S}(C_k) \subseteq \mathsf{S}(P_{k+1})$. Conversely $T_k \subseteq C_k$, so $\mathsf{S}(P_{k+1}) \subseteq \mathsf{S}(C_k)$ and equality holds. Since $P_k \subseteq C_k$, monotonicity follows. If C_k is the unique union of P_k and O_k, applying g gives $\mathsf{S}(C_k) = \mathsf{S}(P_k) \cup \mathsf{S}(O_k)$. A rejected outcome is $\partial \notin \mathcal{X}$ and is absent from O_k, so it changes none of these inclusions. No claim about which member realizes a niche is used. □

Proof of Lemma 1. The two components $Q(C)$ and $F_m(C)$ are disjoint. The first contains exactly the declared score-and-hash winner in each occupied niche. The second is, by definition, exactly the prefix of length $m - |Q(C)|$ of the global order on the remaining candidates. Their union therefore yields the stated necessary and sufficient condition. A previous parent that loses both tests is absent from $T_m(C)$ even though another member of its niche is present. □

5.4 Stage 2 proof of the bridge-chain bound

Proof of Proposition 2. Let $H_0 = E_0$. For $\ell < L - 1$, let $H_{\ell+1}$ be the event that H_ℓ holds, cycle $k + \ell$ generates a member of $B_{\ell+1}$, and selection retains such a member into $P_{k+\ell+1}$. The two ordered conditional bounds and the tower property give

$$\mathbb{P}(H_{\ell+1} \mid H_\ell) \geq a_\ell \rho_\ell.$$

Iterating does not require independence and yields

$$\mathbb{P}(H_{L-1}) \geq \prod_{\ell=0}^{L-2} a_\ell \rho_\ell.$$

Conditional on H_{L-1}, the last generation contains a member of $B_L \subseteq \mathcal{H}_{k+L-1}$ with probability at least a_{L-1}. This proves the generation-level product. Applying the final retention bound ρ_{L-1} proves the acquisition-level product because $P_{k+L} = T_{k+L-1}$. If the charged cycle costs sum to at most B, completion of the certified chain occurs within that budget.

For the operator-specific sufficient condition, choose a retained bridge parent by any fixed measurable rule, for example ascending hash. The designated operator o_ℓ is certainly attempted because the schedule is the full parent–operator Cartesian product. Its kernel mass on $B_{\ell+1}$ is at least a_ℓ and already excludes rejection at ∂. If its accepted output duplicates an earlier child or parent in $B_{\ell+1}$, first-occurrence deduplication still leaves that phenotype in $C_{k+\ell}$; otherwise the new output is added. Thus the operator bound implies the stated generation bound. □

For repeated blocks, condition on each no-hit block boundary. The assumed regeneration property restores the same lower bound p irrespective of the earlier failure history. Applying the history-based survival lemma to blocks gives survival at most $(1-p)^r$ after r blocks and expected charged cost at most $C \sum_{r \geq 0} (1-p)^r = C/p$.

5.5 Dependency audit

- **Conditional independence:** not used. The proof requires history-conditional lower/upper hazard bounds; dependence across offspring or generations is allowed if those bounds remain true after every no-hit history.
- **Markovianity:** not used in the hitting proof. It would only be needed if one chooses a finite-state transition matrix to calculate the hazards. A niche aggregation is valid for that purpose only under lumpability.
- **Mutation bounds:** the quantity α must be a lower bound on reaching the true current hard region conditional on the useful parent being selected. A graph path's existence alone is insufficient; its probability and compute cost matter. The Stage 2 bridge bound instead exposes operator-specific accepted-output mass in $\widetilde{K}_{k,o}$; rejection probability remains in the denominator and no accepted-only renormalization is permitted.

- **Parent retention and scheduling:** a lower bound r must concern availability of a genuinely useful parent at the declared opportunity. Historical presence before a target shift does not establish continued usefulness afterward. The S6-AE parent/operator schedule is deterministic: conditional on P_k, each parent receives all 11 operators. Hence no random parent-selection factor γ may be inserted; a batch reachability bound α_{batch} must be derived under the declared RNG ensemble.
- **Surrogate accuracy:** no accuracy is needed for the abstract survival lemma. To pass from D_k^{gen} to D_k^{acq}, a retention condition connecting $\widehat{h}_k$ and the QD rule to true h_k^{BQ} and future reachability is essential. Perfect current-hardness prediction alone is insufficient when tied candidates have different mutation futures. Without such a bridge, the implementation-level step is OPEN.
- **Capacity and selection:** niche persistence uses $|\mathsf{S}(C_k)| \leq m$ and inclusion of the complete floor $Q(C_k)$. It does not imply bridge persistence. The latter requires the exact elite/global-fill membership condition or another proved retention certificate.
- **Changing hardness:** the bridge theorem allows cycle-indexed kernels and sets, but $B_L \subseteq \mathcal{H}_{k+L-1}$ must refer to true hardness under the entering checkpoint for the final acquisition cycle. Current HGB quality cannot silently replace that condition.
- **Cost normalization:** the comparison assumes equal deterministic cost c per opportunity. If BLADE archive updates, HGB40 inference, or BQ validation change cost, opportunity-count dominance does not imply compute-cost dominance.

5.6 Failure boundary

[**Mathematical status: OPEN for a BLADE-specific dominance claim**] [**Implementation applicability: PARTIAL**] [**Scientific novelty: OPEN**] The proof does not establish that BLADE's conditional hazard is above the fixed-generator or HAC hazard. It fails to produce a BLADE result at the exact step where one would need verified bounds on r and α_{batch} (or a direct generation/acquisition hazard bound) and a comparator upper bound p_G, all in the same compute units. The counterexamples show these inequalities can reverse. No unproved transition from a structural-memory narrative to hazard dominance is made.

6 Adversarial finite-state checks

The examples below are exact toy models. They test universal statements and assumptions; none is asserted to be the BLADE-NCO implementation. Their tags report mathematical status; implementation applicability is [**Implementation applicability: NOT APPLICABLE as a universal guarantee**], and novelty is [**Scientific novelty: NOT CLAIMED**].

6.1 Stale memory after a target shift

[**Mathematical status: DISPROVED**] for the claim that historical memory cannot hurt. Take $\mathcal{X} = \{a, b\}$. Before a change the hard region is $\mathcal{H}_0 = \{a\}$; just before the first measured proposal it shifts to $\mathcal{H}_i = \{b\}$ for all $i \geq 1$. The inherited population contains only a. The memory policy mutates from a to a with probability 0.99 and to b with probability 0.01, retaining a after a miss. Each proposal costs one unit. Its hitting time is geometric with success probability 0.01, so

$\mathbb{E}[\tau_M] = 100$ and

$$\mathbb{P}(\tau_M \leq B) = 1 - 0.99^B.$$

A fixed generator that independently samples a or b uniformly every proposal has $\mathbb{E}[\tau_G] = 2$ and success probability $1 - 0.5^B$. Thus the memory policy is strictly worse at every positive integer budget. The target has undergone a single non-stationary shift; the failure is stale structural support, not an independence artifact.

6.2 Gradient adaptation can beat structural mutation

[**Mathematical status: DISPROVED**] for any universal claim that structural mutation outperforms HAC-style gradient adaptation. Let $\mathcal{X} = \{0, 1, \ldots, L\}$ and $\mathcal{H} = \{L\}$. A structural mutation from $i < L$ advances to $i+1$ with probability ε and otherwise stays at i. Starting at 0, its expected number of unit-cost proposals to hit L is L/ε. A gradient-informed comparator that advances deterministically one step per unit-cost proposal hits in L proposals. For any $0 < \varepsilon < 1$, the structural policy is slower. This is a toy landscape with an informative gradient, not a model of the actual HAC implementation; it proves that operator labels alone imply no ranking.

6.3 Diversity can allocate effort away from the target

[**Mathematical status: DISPROVED**] for the claim that increased diversity always improves first hitting time. Let $\mathcal{X} = \{s, u, g\}$, target $\mathcal{H} = \{g\}$, and one proposal cost one unit. The diverse parent set contains distinct niches s and u; parent selection is uniform. Mutation from s reaches g surely, whereas mutation from u stays at u. Hence the per-proposal hit probability is $1/2$ and expected hitting cost is 2. A focused policy that selects s every time hits in cost 1. Preserving the irrelevant niche does not help this target and reduces access to the useful parent.

6.4 Perfect current-hardness prediction need not preserve a bridge

[**Mathematical status: DISPROVED**] for the claim that perfect prediction of current BQ hardness makes score-elite retention preserve future mutation reachability. Consider states $\{a, b, g\}$ and current target $\mathcal{H} = \{g\}$; neither a nor b is itself a hit, and their current true hardness is equal: $h(a) = h(b) < \kappa$. From a, the next scheduled mutation reaches g surely; from b, mutation stays at b surely. Let the surrogate be perfect on current hardness, so $\widehat{h}(a) = h(a) = h(b) = \widehat{h}(b)$. A one-slot score-elite update must still break the tie. If its deterministic tie rule retains b, it discards a and makes g unreachable. Thus current-hardness accuracy, even perfection, does not identify equal-hardness candidates with different mutation futures. A reachability-aware rule could retain a, but perfect current-hardness prediction alone need not do so. This is not a claim that the observed S6-AE hash tie-break selected such a pair; it is an exact obstruction to the proposed bridge.

6.5 Monotone niche occupancy with progressive bridge loss

[**Mathematical status: PROVED counterexample**] to the claim that monotone niche occupancy preserves future hard-region reachability. Fix population capacity m and an integer

$2 \le L \le \min\{K-1, m-1\}$. Use occupied niches $0, 1, \ldots, L$. Initially P_0 contains a bridge a_i in niche i for each $1 \le i \le L$, together with $m - L$ distinct fillers $f_1, \ldots, f_{m-L}$ in niche 0.

At cycle $i-1$, one designated operator applied to a_i produces a valid decoy d_i in the same niche i with probability one. Every other scheduled attempt returns the rejection symbol ∂; no attempt is resampled. Set the current true hardness and a perfect current predictor so that

$$h(a_i) = \widehat{h}(a_i) = 1, \qquad h(d_i) = \widehat{h}(d_i) = 2, \qquad h(f_j) = \widehat{h}(f_j) = 3,$$

with deterministic distinct hash order where needed. Choose the current hard threshold above 3, so none of these candidates is a current hit.

In niche i, d_i replaces a_i as the score elite. Niche 0 contributes one filler to the elite floor, and the remaining $m - L - 1$ fillers occupy all global-fill slots. Thus a_i fails both tests in Lemma 1 and is discarded. All as-yet-unprocessed a_j remain the sole candidates in their niches and therefore survive. Inductively, after cycle t the selected population still occupies exactly $\{0, 1, \ldots, L\}$, while the number of retained bridges has fallen from L to $L - t$.

After cycle $L-1$, let the future true-hard region be $\{g\}$. A future designated operator reaches g surely from any a_i and cannot reach it from any d_i or f_j. The selected population now contains no a_i, so its future hit probability is zero, even though its niche occupancy never decreased. A population retaining any discarded a_i would hit surely. The failure is not surrogate error: the predictor is perfect for current hardness. It is a mismatch between current-hardness quality and future-reachability value.

This construction includes the complete elite floor and global score fill, not merely a one-slot archive. Taking, for example, $m = 10{,}000$ and $L \le 999$ respects the S6-AE capacity ratio. It is a formal counterexample, not evidence that these states or score relations occurred in S6-AE.

6.6 Niche aggregation can destroy the Markov property

[**Mathematical status: DISPROVED**] for the assertion that a niche label alone is a Markov state. Let full states x_1, x_2 both map to niche A, while z, w map to distinct niches Z, W. From x_1, the next state is z with probability 1; from x_2, the next state is w with probability 1. Both full states have the same current aggregate A, but the next aggregate is respectively Z or W. The transition probability out of aggregate A is therefore not determined by the aggregate alone (strong lumpability fails). Therefore niche aggregation is not Markov uniformly over initial laws. For a specially chosen initial distribution, an averaged transition might nevertheless be Markov; that weaker distribution-dependent claim would require the conditional within-A mixture to remain history-independent. The example does not disprove every such weak aggregation. A proof using only niche counts is invalid unless it proves the relevant strong or distribution-dependent condition, or augments the state with the hidden structural/solver information.

7 Research assessment

Strongest candidate. The strongest Stage 2 candidate is Proposition 2. It turns a multicycle structural path into an auditable finite-budget bound using operator-specific accepted-transition masses and explicit selection-retention certificates. Unlike Proposition 3, it does not begin by

assuming an unexplained overall hazard advantage. Proposition 1 supplies a verified structural invariant, but the progressive-loss counterexample shows why that invariant cannot supply the bridge-retention premise by itself.

Limitations and novelty. The product/conditioning argument and archive occupancy fact are standard and are not claimed as new theory. Scientific content would have to come from deriving non-vacuous FOGA-11 kernel bounds, proving that current-HGB elites retain future-relevant bridges despite BQ drift, and obtaining a strict compute-normalized separation from specified comparators. The inspected implementation verifies the deterministic operator schedule, rejection without resampling, population update, capacity relation, and $P = T$, but not those probabilistic or score–reachability premises. The result therefore does not assert a BLADE advantage.

Stage 2 verifies the S6-AE batch schedule and population update rule, but not the required probabilistic hazard bounds, a scientific hard-region threshold, or a matched HAC transition/cost law. Therefore the mathematical comparison remains conditional, its BLADE applicability remains partial, and its novelty remains open.

Testability. The structural persistence proposition is checkable directly from the selector definition and existing artifacts. The bridge theorem would require certified support or lower bounds for accepted operator transitions between declared bridge sets and proof that the selector retains those sets. The inspected artifacts record proposals, rejections, scores, and memberships, but a single frozen trajectory cannot establish history-uniform kernel bounds.

[Implementation applicability: OPEN]

No new evaluations, training, or evolutionary runs were performed for this note.

Unresolved issues. The decisive open issue is whether current-hardness selection preserves representatives with high future-reachability value often enough to complete a nontrivial bridge chain. Exact operator laws under a declared ensemble, score–bridge alignment, target sets, and matched fixed-generator/reset/HAC cost models remain necessary before a BLADE-specific comparison theorem can be claimed.

8 Stopping point

Stage 1 formalization, Stage 1.5 corrections, and the focused Stage 2 persistence/reachability investigation are recorded. No experiment, algorithm change, Oracle/BQ evaluation, retraining, adaptation, evolution, or broader Stage 3 development is included.

References

[1] D.-C. Dang, T. Jansen, and P. K. Lehre, "Populations Can Be Essential in Tracking Dynamic Optima," *Algorithmica*, vol. 78, no. 2, pp. 660–680, 2017. doi: 10.1007/s00453-016-0187-y.

[2] P. S. Oliveto and C. Zarges, "Analysis of diversity mechanisms for optimisation in dynamic environments with low frequencies of change," *Theoretical Computer Science*, vol. 561, part A, pp. 37–56, 2015. doi: 10.1016/j.tcs.2014.10.028.

[3] J.-B. Mouret and J. Clune, "Illuminating Search Spaces by Mapping Elites," arXiv:1504.04909, 2015.

[4] J. K. Pugh, L. B. Soros, and K. O. Stanley, "Quality Diversity: A New Frontier for Evolutionary Computation," *Frontiers in Robotics and AI*, vol. 3, art. 40, 2016. doi: 10.3389/frobt.2016.00040.

[5] J. Bossek and D. Sudholt, "Runtime Analysis of Quality Diversity Algorithms," *Algorithmica*, vol. 86, pp. 3252–3283, 2024. doi: 10.1007/s00453-024-01254-z.

[6] C. Qian, K. Xue, and R.-J. Wang, "Quality-Diversity Algorithms Can Provably Be Helpful for Optimization," arXiv:2401.10539, 2024.